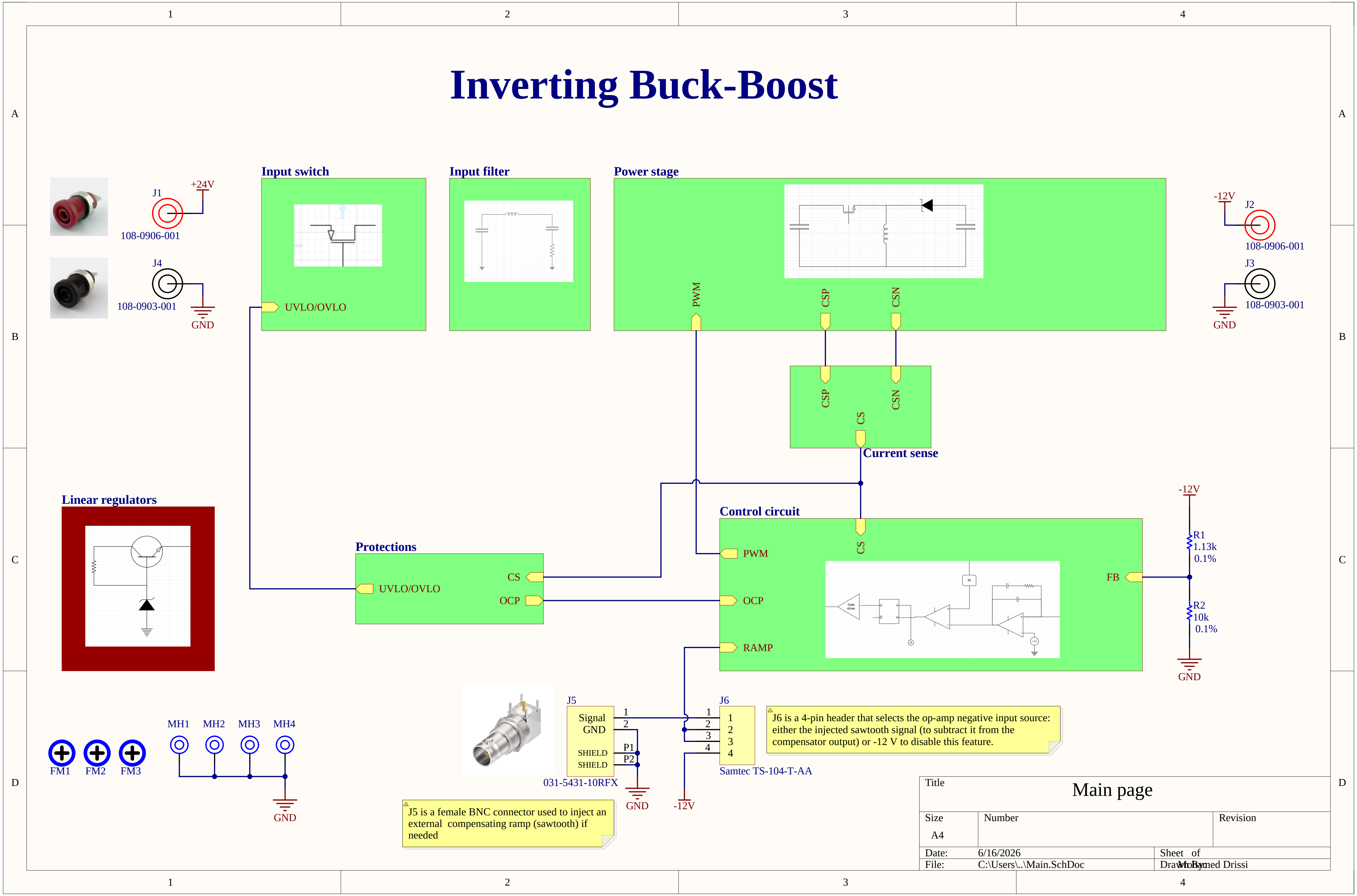
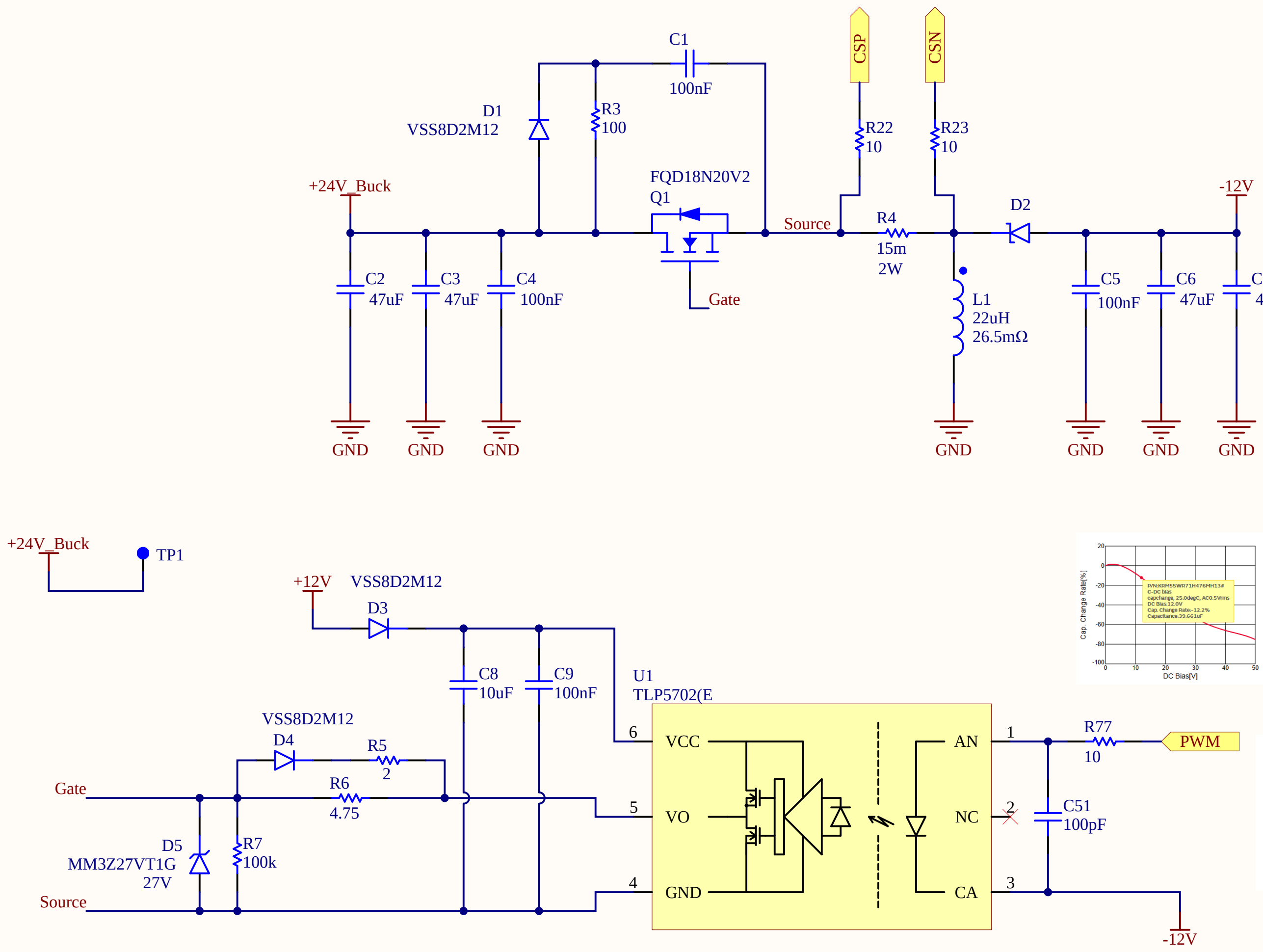
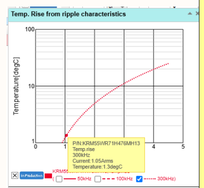
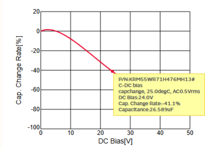
Inverting Buck-Boost

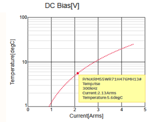


Power stage









ANALYSIS

Vin=24V | Vout=-12V Iout=3A | fs=300kHz | D=0.333 | eff=88%

INDUCTOR — SRP1245A-220M (22uH)

IL_avg = Iin + Iout = 1.705 + 3 = 4.705A
ΔIL = Vin×D / (fs×L) = 1.21A
IL_peak = IL_avg + ΔIL/2 = 5.31A
IL_ripple_ratio = 1.21/4.705 = 25.7%
L_crit = $(D')^2$×R / 2fs = 2.97uH
CCM confirmed: L=22uH >> Lcrit=2.97uH ✓

MOSFET — FQD18N20V2 (200 V, 15 A, 140 mΩ)

Vds_stress = Vin + Vout = 24+12 = 36V
Vds_rated = 36×1.2 = 43.2V → 600V ✓ (20% derating)
Id_rated ≥ IL_peak = 5.31A → 15A ✓
ID_rms = IL_avg × √D = 2.72A
P_cond = ID_rms² × Rds(on)×1.3 = 2.72^2×0.637 = 4.71W (30% derating)
P_sw = 0.5×Vds×IL_peak×(tr+tf)×fs = 0.5×36×5.31×20n×300k = 0.57W
P_total = 4.71 + 0.57 = 5.28W

DIODE — STPS10L60S (60V, 10A, Vf=0.4V, DPAK)

Vr_stress = Vin + Vout = 36V
Vr_rated = 36×1.2 = 43.2V → 60V ✓
ID_avg = Iout = 3A
ID_peak = IL_peak = 5.31A
P_diode = Vf×Iout×(1-D) = 0.4×3×0.667 = 0.8W

OUTPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47uF,50V,X7R)

C_min = Iout×D / (fs×ΔVout) = 27.7uF
C_nominal = 2×47 = 94uF
C_derated @ 12V = 2×39.6 = 79.2uF > 27.7uF ✓
ΔVout_target= 1%×12 = 120mV
ESR_max = 0.5×ΔVout/ΔIL = 49.6mΩ
(the ESR can contribute max by a 50% to the total allowed ripple)
ESR_eff = 4mΩ/2 = 2mΩ << 49.6mΩ ✓
IC_rms = Iout×√(D/D') = 2.12A
IC_rms/cap = 2.12/2 = 1.06A ✓

INPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47uF,50V,X7R)

C_min = Iout×D/(fs×ΔVin×D') = 20.8uF
C_nominal = 2×47 = 94uF
C_derated @ 24V = 2× 26.6uF = 53.2uF > 20.8uF ✓
ΔVin_target = 1%×24 = 240mV
ESR_max = 0.5×ΔVin/IL_peak = 22.6mΩ
ICin_rms = IL×√(D×D') = 2.22A ✓
Vrated = 32×1.5 = 48V → 50V ✓
Cin impedance Zcin = 1/(2π×fs×Cin) + ESRcin = 0.413Ω
Cdamp impedance Zcdamp = 1/(2π×fs×Cdamp) + ESRdamp = 0.013Ω
IRMS Total Input Caps (Cin +Cdamp) = IL×√(D×D') = 2.22A
IRMS Cdamp = Zcdamp / (Zcdamp +Zcin) = 2.16A✓

POWER BUDGET

P_mosfet = 5.28W
P_diode = 0.80W
P_total = 6.08W
η_estimated = Pout/(Pout+Ploss) = 36/(36+6.08) = 85.6%

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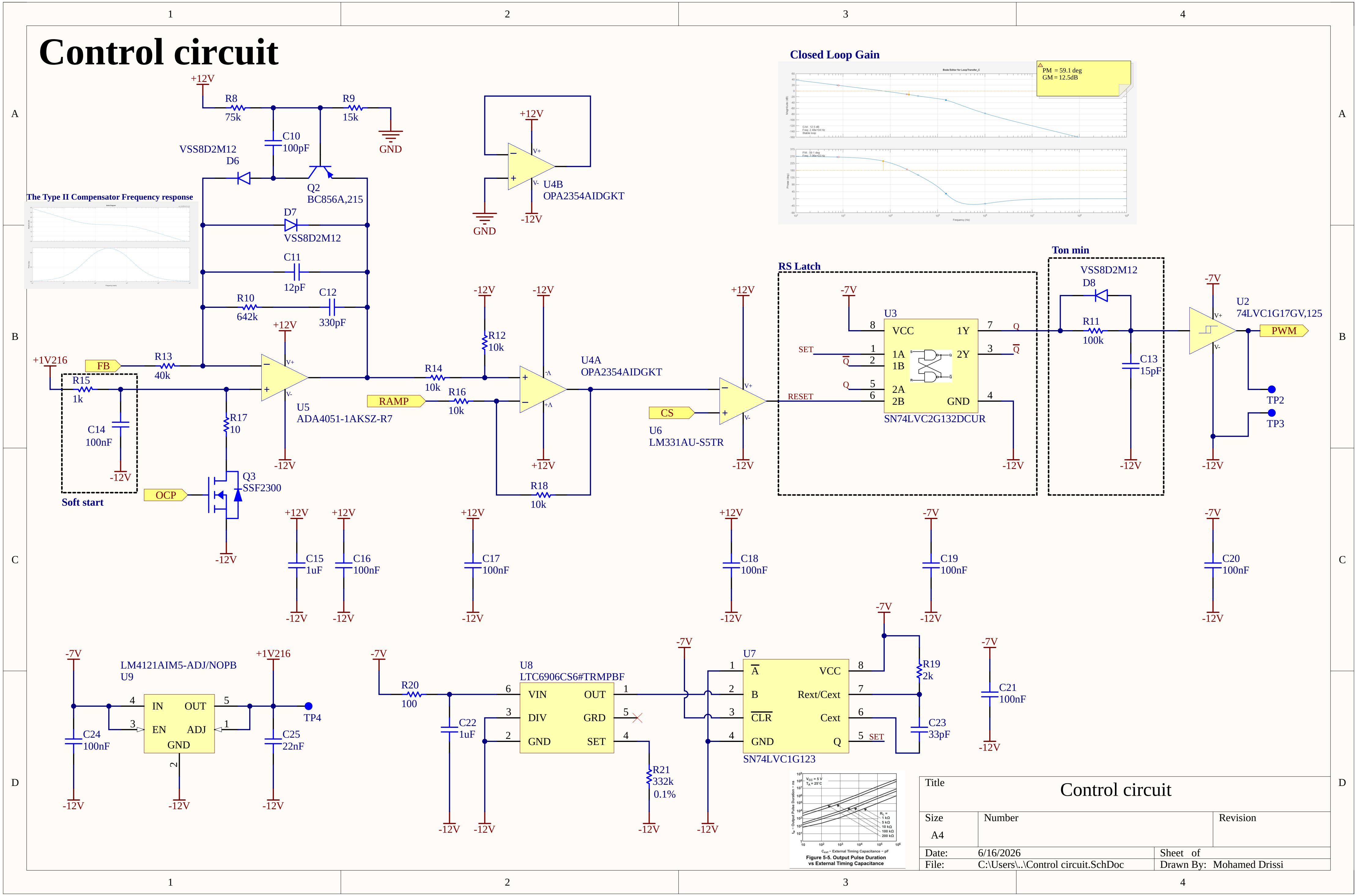
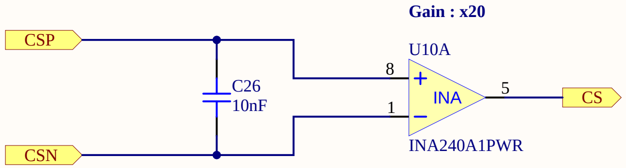
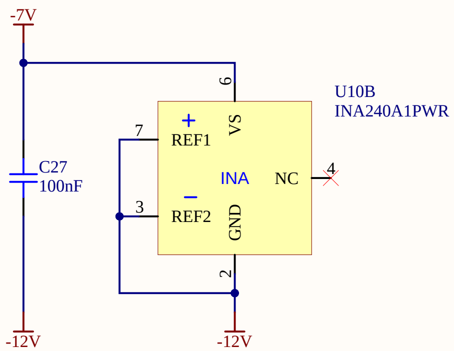
Control circuit



Current sense



Total current gain = 0.015 x 20 = 0.3



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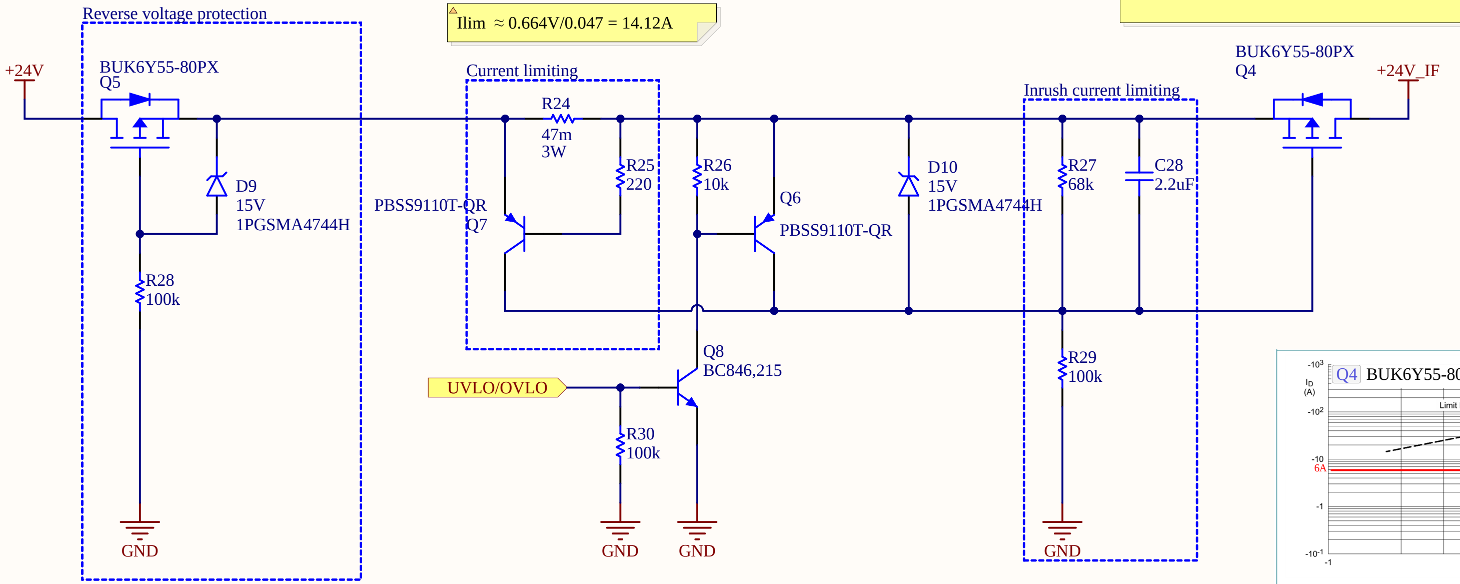
Input switch

STATIC POWER DISSIPATION for Q4

$P = I^2 \times RDS(on)$
$= (1.5\ A)^2 \times 0.055\ \Omega$
$= 0.124\ W$

Maximum static power dissipation: 124 mW
Estimated junction-to-case temperature rise: 0.12 °C
($R\theta J\text{-}MB = 1$ °C/W)

Ilim ≈0.664V/0.047 = 14.12A



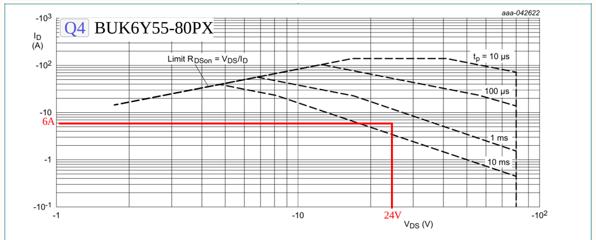


Fig. 3. Safe operating area; junction to mounting base; continuous and peak drain currents as a function of drain-source voltage

Cload = Cf +Cdamp + Cin
= 5.45u + 53.3u + 220u = 278.75u ≈300u
Δt = C_load × V_in / I_inrush_max
Δt = 300uF × 24V / 6A = 1.2ms (The MOSFET Q4 can withstand pulses longer than 1 ms. The SOA analysis is a conservative method for verifying whether the transistor can support the inrush current while operating in the linear region. In this analysis, the current is assumed to be a square wave and the VDS is assumed to remain constant at 24 V throughout the entire 1.2 ms inrush period. Based on these conservative assumptions, the MOSFET can be considered suitable for this application.)
based on LTSpice simulation a cap value of 2.2uF C28
in combination with 100K R29 we can achieve approx Iinr = 5.8A within 800us which is sufficient and Q4 can withstand
Vgate(steady state) = Vin×R29/(R27+R29) = 24×100k/(68k+100k)=24×0.595 = 14.3V
VGS,ss = Vgate,ss − Vin = 14.3 − 24 = 9.7V more than Vgs threshold and less than Vgs max of

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Input filter

INPUT EMI FILTER : Design Notes

Standard: EN 55032:2015 Class B — Conducted Emissions
Limit @ 300kHz = 56 dBµV (quasi-peak)
Frequency range: 150kHz – 30MHz
SPECIFICATIONS:
Vin = 24V | Vout = -12V | Iout = 3A | fs = 300kHz | D = 0.333 | Cin = 53.2uF (derated @24V)

STEP 1 : NOISE LEVEL ESTIMATE

First harmonic amplitude:
$I1 = (2\times IL\times \sin(\pi\times D)) / \pi = (2\times 4.705\times \sin(\pi\times 0.333)) / \pi = 2.60A$
Noise voltage across Cin at fs:
$Vnoise = I1 / (2\pi\times fs\times Cin) = 2.60 / (2\pi\times 300k\times 53.2u) = 25.9mV = 25.9\times 10^3\ \mu V$
Estimated noise level:
$|Noise|dB\mu V = 20\log(25.9\times 10^3) = 88.3\ dB\mu V$
Required attenuation:
$Att = |Noise|dB\mu V - Vmax = 88.3 - 56 = 32.3\ dB$
Adding 10dB safety margin:
$Att\_target = 32.3 + 10 = 42.3\ dB$

STEP 2 : INDUCTOR SELECTION

Lf = 10uH → Selected: 10uH, 12A, SMD

STEP 3 : CAPACITOR SELECTION

Resonance one decade below fs:
$Cf = Cin / (Cin\times Lf\times (2\pi\times fs/10)^2 - 1) = 53.2u / (53.2u\times 10u\times (2\pi\times 30k)^2 - 1 = 3\mu F$
Required attenuation:
$Cf = 1/Lf \times (10\verb|^|(Att/40)/(2\pi\times fs))^2 = 5\mu F$
Select higher value: Cf = max(5µF,3µF) = 5µF→ 10µF selected derated to 5.45µF @24V
NOTE: Cf is a SEPARATE dedicated filter capacitor distinct from Cin.

$fc\_filter = 1/(2\pi\times\sqrt{Lf\times Cf}) = 1/(2\pi\times\sqrt{10u\times 5.45\mu F}) = 21.6kHz$
Attenuation verification:
$A = 40\times\log(fs/fc\_filter) = 40\times\log(300k/21.6k) = 45.7dB >> 42.3dB$ ✓

STEP 4 : DAMPING CAPACITOR

$Cd \geq 4\times Cin = 4\times 53.2u = 212.8uF$ → 220uF selected

Required ESRd:
$ESRd \leq \sqrt{Lf/Cin} = \sqrt{10u/53.2u} = 0.434\Omega$

Damping resistor power:
$P\_Rdamp \approx 0.5\times\Delta Iin^2\times Rdamp = 0.5\times(1.21)^2\times 1.5 = 0.8W$
Selected Cd: 220uF 35V polymer electrolytic ESR_polymer ≈ 11mΩ < 434mΩ ✓
NOTE: Polymer ESR too low, added series resistor Rd = 0.4Ω in series with Cd to meet ESRd target
Voltage rating: Vin_max×1.25 = 32×1.25 = 40V → 50V ✓
Cin impedance Zcin = 1/(2π×fs×Cin) + ESRcin = 0.413Ω
Cdamp impedance Zcdamp = 1/(2π×fs×Cdamp) + ESRdamp = 0.013Ω
IRMS Total Input Caps (Cin +Cdamp) = IL×√(D×D') = 2.22A
IRMS Cdamp (current divider) = Zcin / (Zcdamp +Zcin) = 0.062A✓

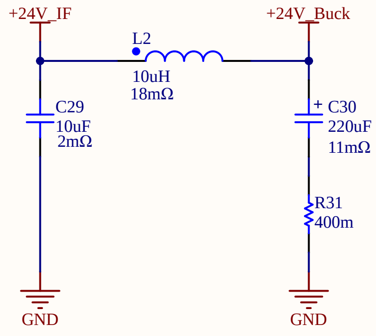
STEP 4 : MIDDLEBROOK CRITERION

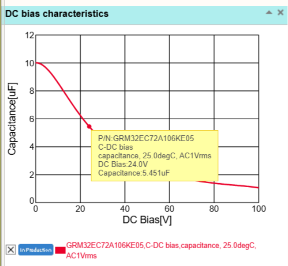
$Zin\_conv = Vin^2\times\eta/Pout = 576\times 0.88/36 = 14.1\Omega$
Full frequency spectrum verified using matlab: see Plot

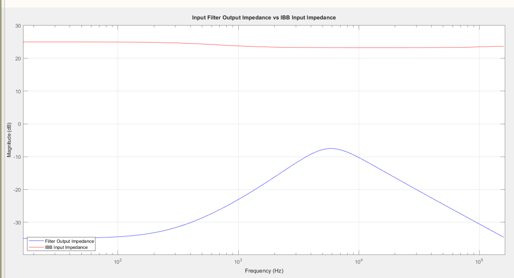
EMC STANDARD : EN 55032:2015

EN 55032 "Electromagnetic Compatibility of Multimedia Equipment" . It defines conducted emission limits on the power port for Class B equipment (residential environment):
150kHz – 500kHz : 56–66 dBµV (quasi-peak)
46–56 dBµV (average)
500kHz – 30MHz : 56 dBµV (quasi-peak)
46 dBµV (average)
Worst case for this design at fs=300kHz: 56 dBµV QP.

Reference : Conducted EMI Characteristics And Mitigation Technique
https://www.ti.com/lit/an/snva489c/snva489c.pdf







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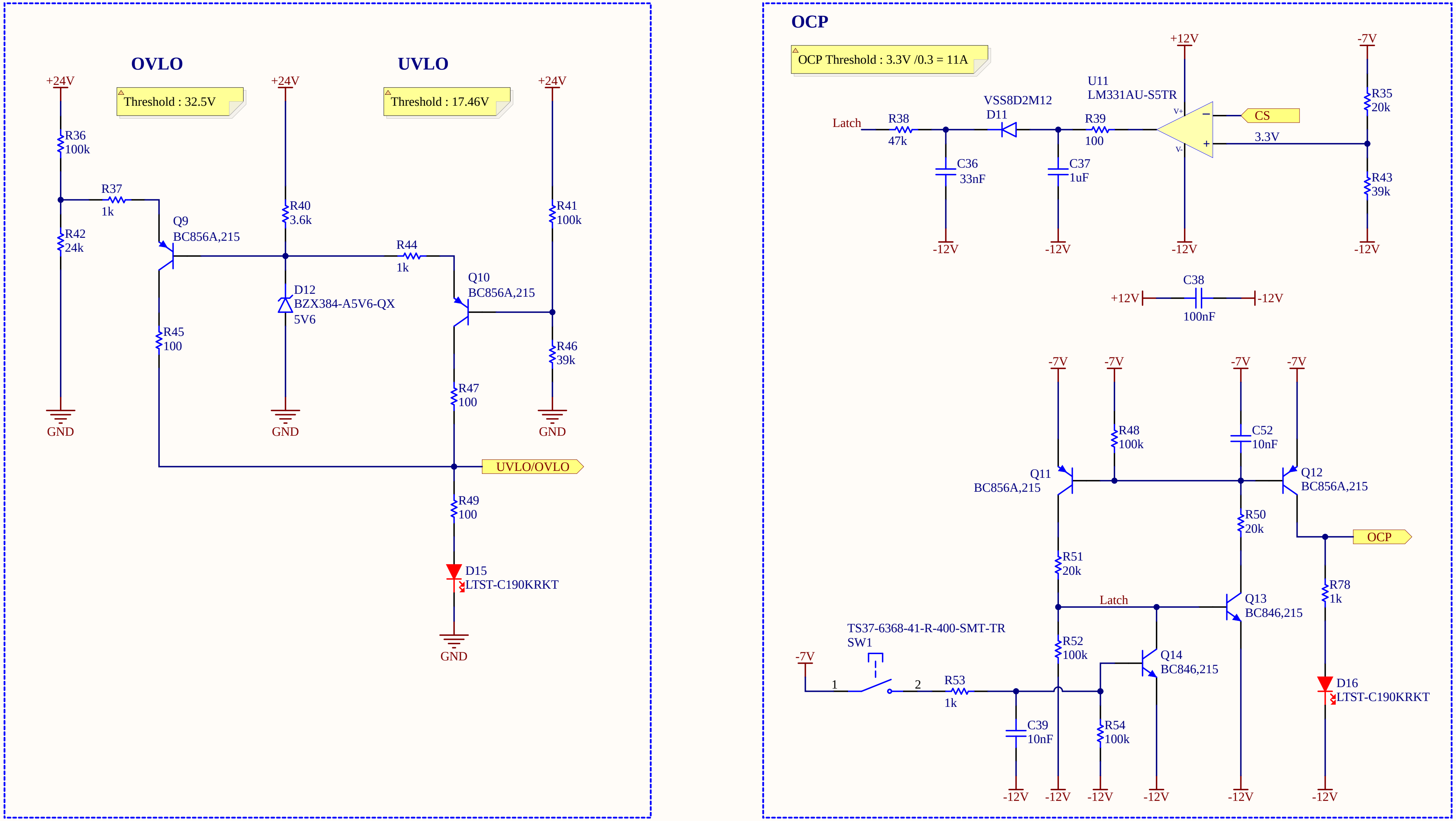
Protections



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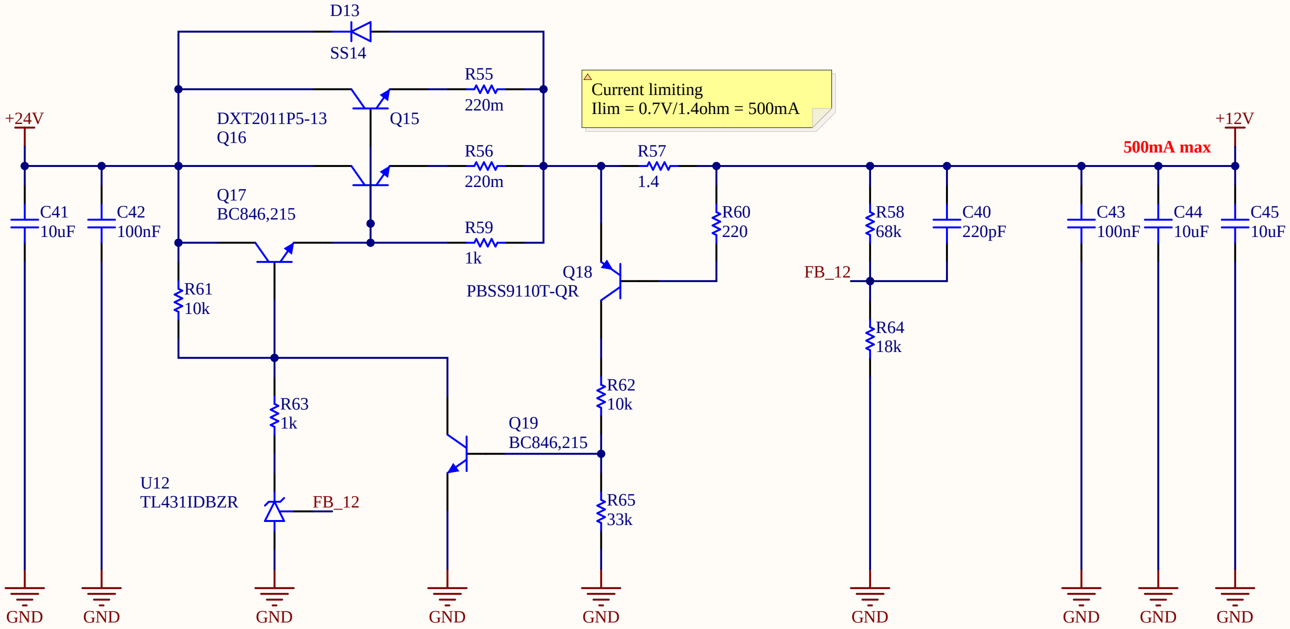
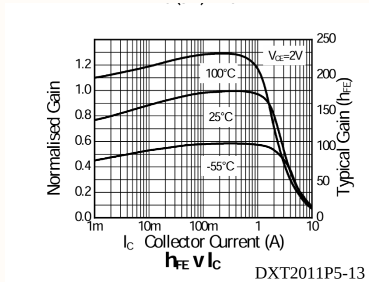
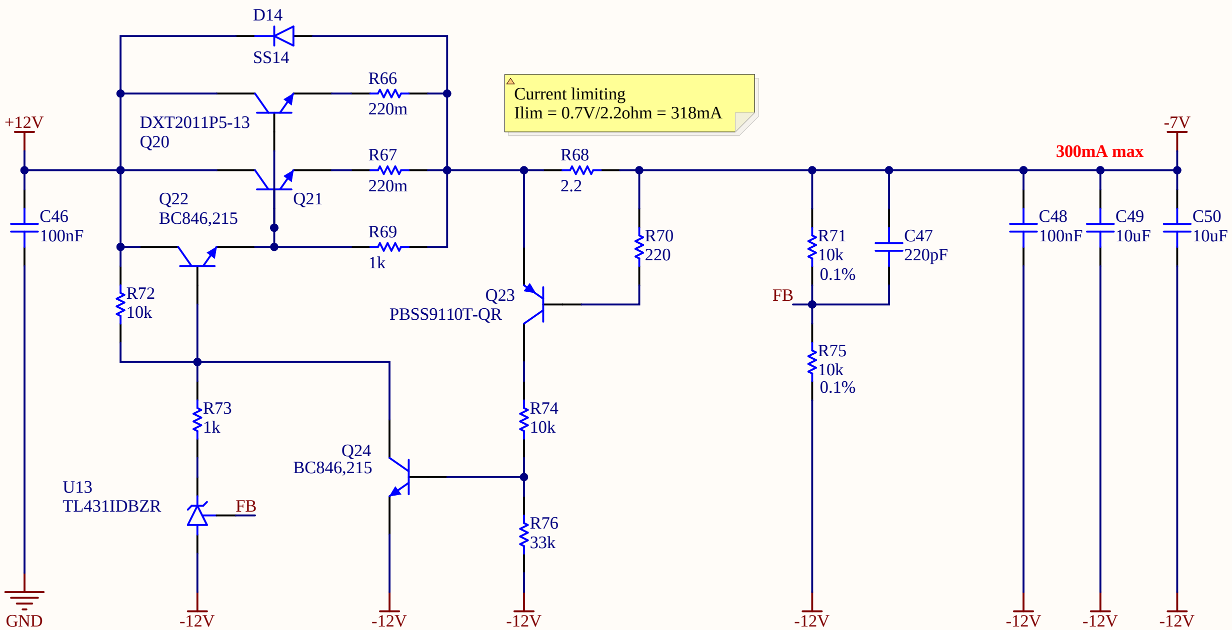
Linear regulators

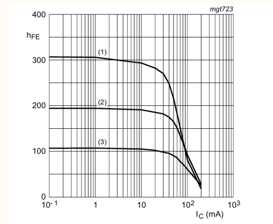


To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable hFE. When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet The base-emitter resistor (R59 shown in the 12V regulator) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.



DXT2011P5-13





BC846,215

ig. 2. Group A: DC current gain as a function of collector current; typical values

DXT2011PS can dissipate 3W if the device mounted with the exposed collector pad on 50mm x 50mm 2oz copper that is on a single-sided 1.6mm FR4 PCB

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